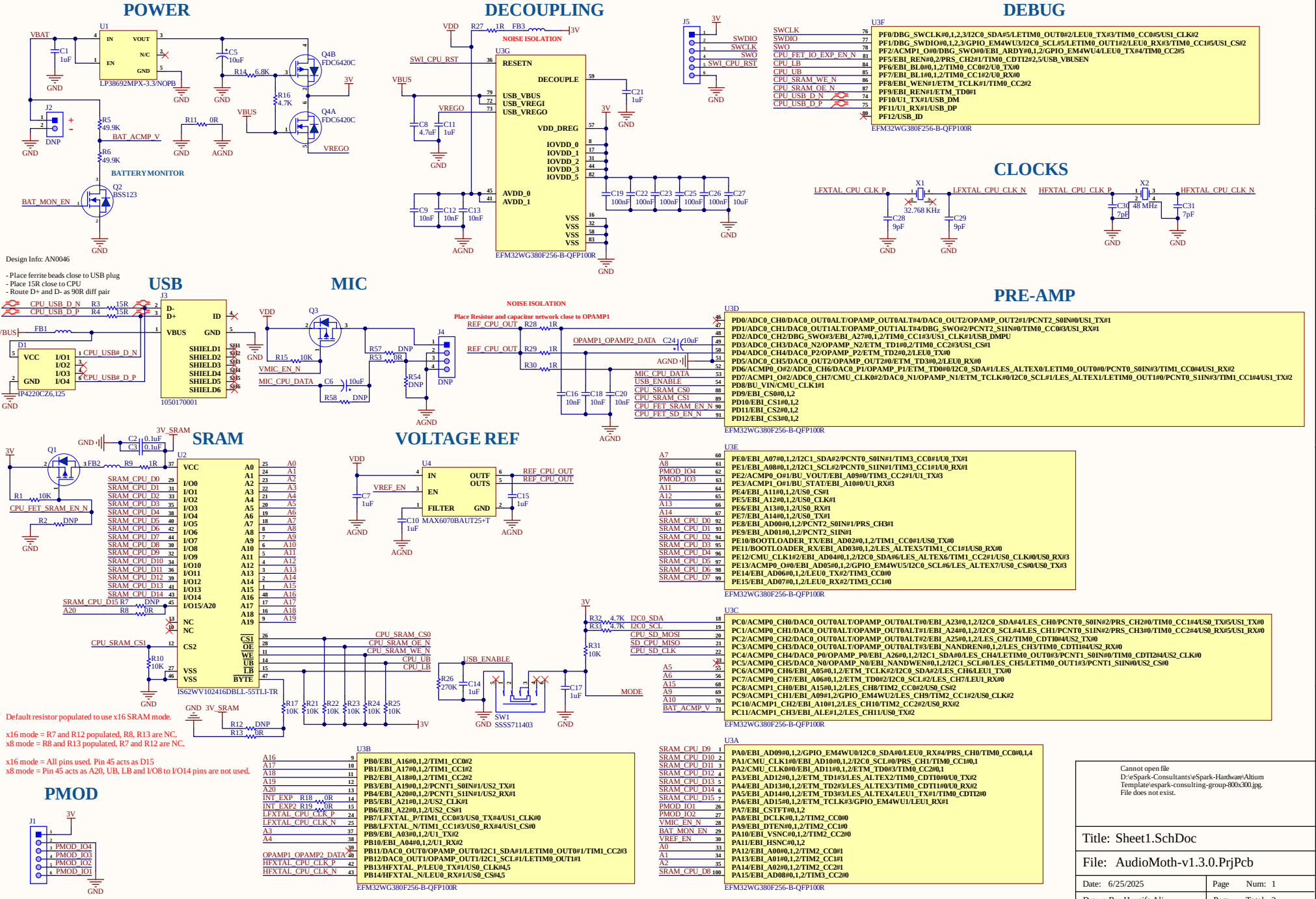
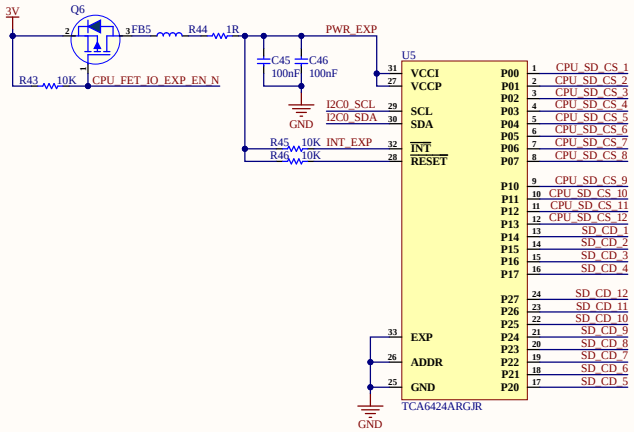
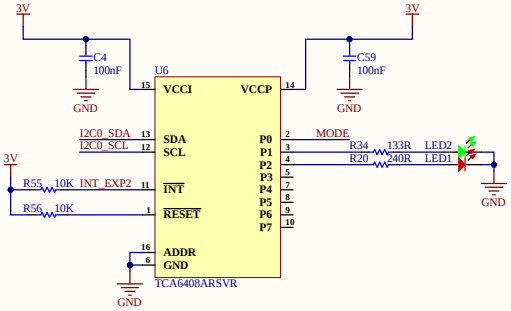

POWER
DECOUPLING
DEBUG
CLOCKS
USB
MIC
PRE-AMP
SRAM
VOLTAGE REF
PMOD
Design Info: AN0046
- Place ferrite beads close to USB plug
- Place 15R close to CPU
- Route D+ and D- as 90R diff pair
NOISE ISOLATION
Place Resistor and capacitor network close to OPAMP1
Default resistor populated to use x16 SRAM mode.
x16 mode = R7 and R12 populated, R8, R13 are NC.
x8 mode = R8 and R13 populated, R7 and R12 are NC.
x16 mode = All pins used. Pin 45 acts as D15
x8 mode = Pin 45 acts as A20, UB, LB and I/O8 to I/O14 pins are not used.
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.
Title: Sheet1.SchDoc
File: AudioMoth-v1.3.0.PrjPcb
Date: 6/25/2025
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 2

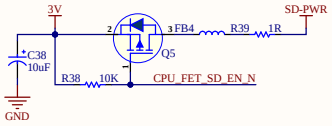
IO EXPANDER 1



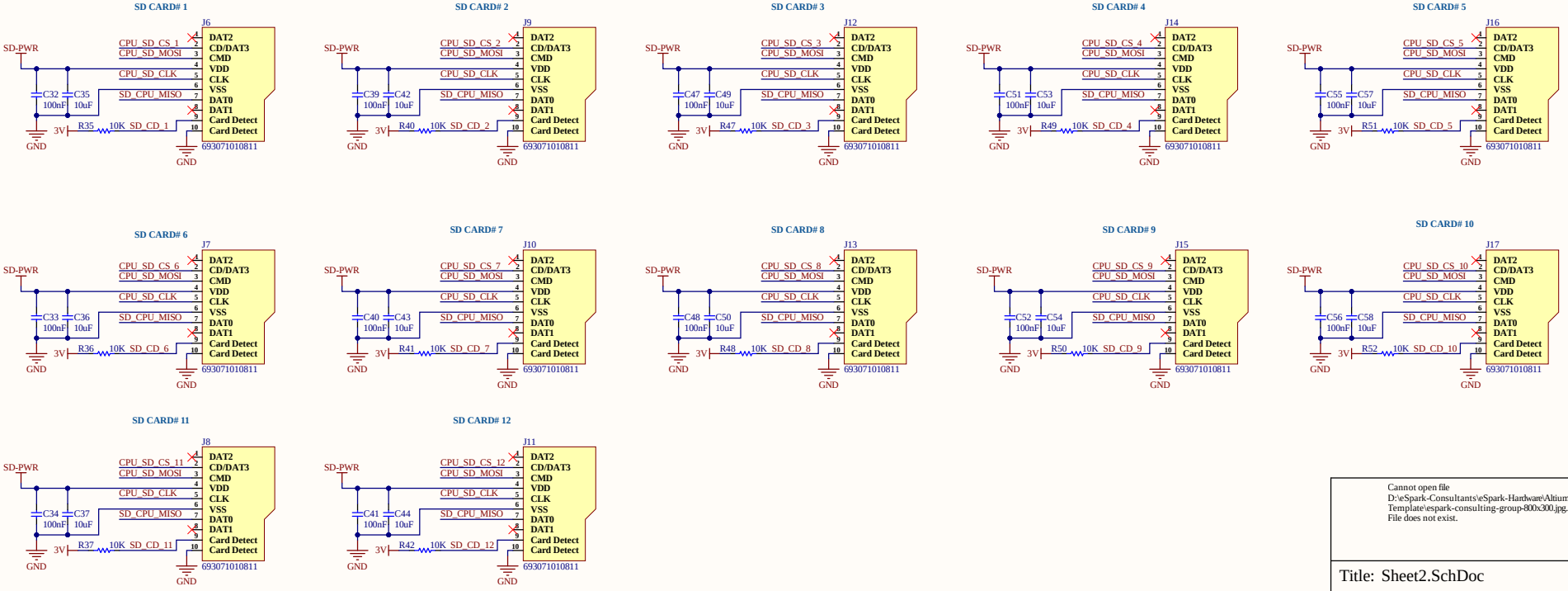
IO EXPANDER 2



SD POWER SWITCH

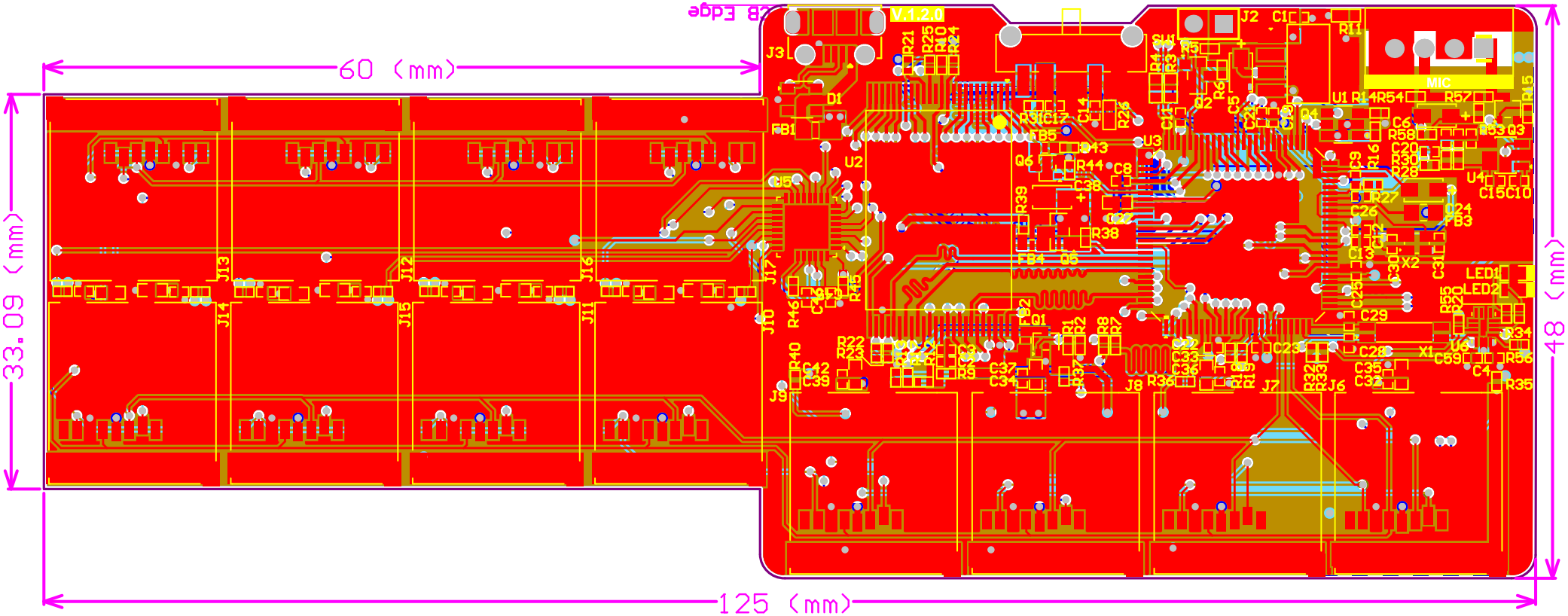


SD CARDS



Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Sheet2.SchDoc	
File: AudioMoth-v1.3.0.PrjPcb	
Date: 6/25/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 2


60 (mm)
33.09 (mm)
125 (mm)
48 (mm)
V.1.2.0
MIC
J1
J2
J3
J4
J5
J6
J7
J8
J9
J10
J11
J12
J13
J14
J15
J16
J17
U1
U2
U3
U4
U5
U6
LED1
LED2
X1
X2